

Sheth l.u.j. And sir m.v. college of arts science and commerce

Cyber & Information Security Practical No. 2

Aim:

Implement the RSA algorithm for public-key encryption and decryption, and explore its properties and security considerations.

RSA (Rivest–Shamir–Adleman) is a public-key cryptography algorithm used to securely encrypt and decrypt data using two keys: a public key and a private key.

Public Key: The public key is used to encrypt the message and can be shared with anyone.

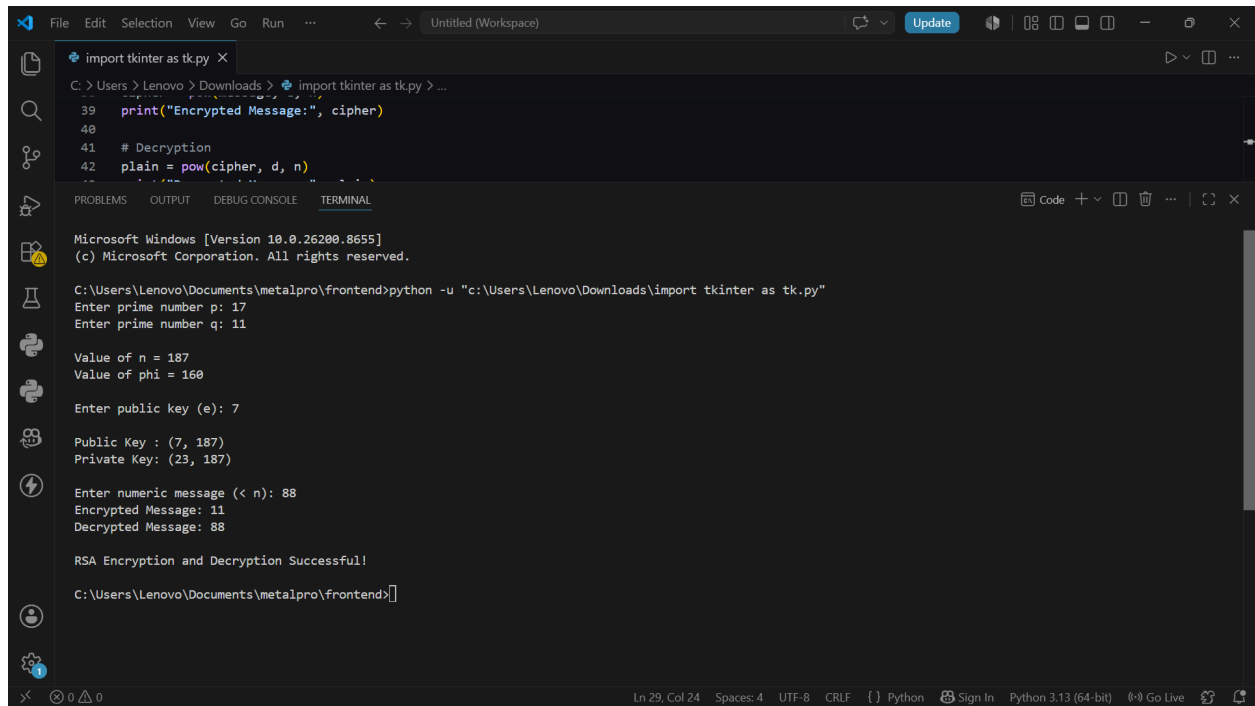
Private Key: The private key is used to decrypt the encrypted message and must be kept secret.

Encryption: Encryption converts the original message into ciphertext using the public key.

Decryption: Decryption converts the ciphertext back into the original message using the private key.

CLI

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```
import tkinter as tk.py
C:\Users\Lenovo\Downloads> import tkinter as tk.py > ...
39 print("Encrypted Message:", cipher)
40
41 # Decryption
42 plain = pow(cipher, d, n)
...
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Lenovo\Documents\metalpro\frontend>python -u "c:\Users\Lenovo\Downloads\import tkinter as tk.py"
Enter prime number p: 17
Enter prime number q: 11

Value of n = 187
Value of phi = 160

Enter public key (e): 7

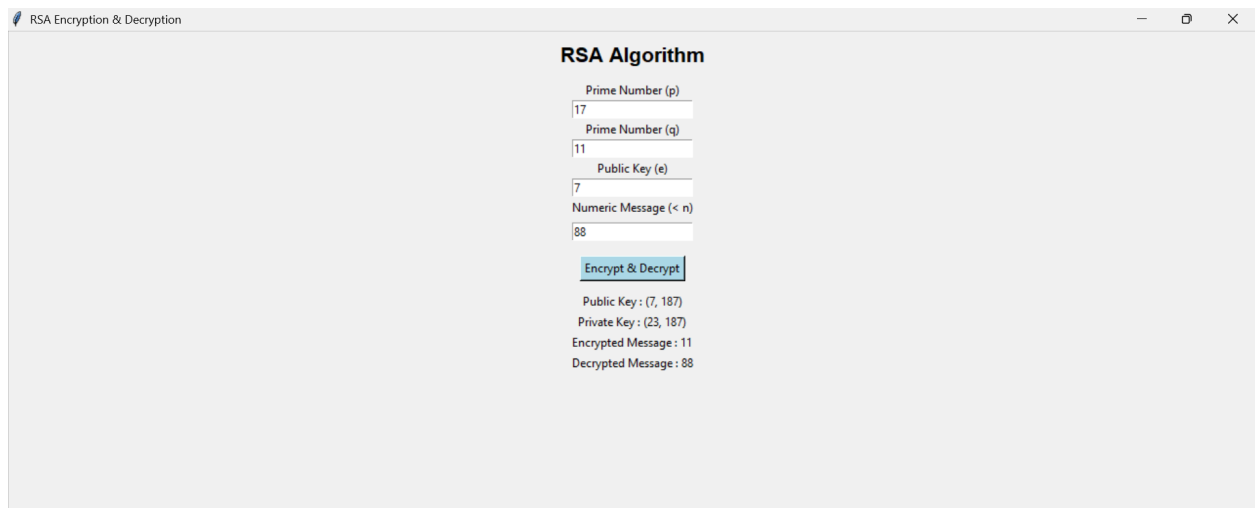
Public Key : (7, 187)
Private Key : (23, 187)

Enter numeric message (< n): 88
Encrypted Message: 11
Decrypted Message: 88

RSA Encryption and Decryption Successful!

C:\Users\Lenovo\Documents\metalpro\frontend>
```

Gui



RSA Algorithm

Prime Number (p)
17

Prime Number (q)
11

Public Key (e)
7

Numeric Message (< n)
88

Encrypt & Decrypt

Public Key : (7, 187)
Private Key : (23, 187)
Encrypted Message : 11
Decrypted Message : 88

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